ENDPOINT LOCALIZATION FOR TYPE III AND TYPE IV SALEM–BOYD INTERLACING FAMILIES

KEO

Abstract. McKee and Smyth proved that every Salem number admits at least one, generally non-unique, interlacing witness of Types I–IV. We study relative sequential closure in $(1, \infty)$, with emphasis on the moving-pole Types III and IV. Type I is closed in the interior, while an exact CS-to-CC transformation puts Type II in a closed CC chart.

For Type III we prove closure away from collision between the output and the denominator's exterior root, as well as on the positive-residue and bounded-denominator-degree strata. Every remaining collision has unbounded-degree Salem denominator roots, and its residue is strictly linearly comparable with the pole–output gap; in particular, a higher-order sublinear collision is impossible. We also give an explicit Type III witness showing that the Type II Weyl descent cannot be iterated inside any fixed integral Laurent lattice.

For Type IV an exact nonnegative defect identity yields the universal bound $c < 1/2$, a quantitative pole–output separation, and a three-stratum subsequential compactification: finite separation, zero-residue collision, or half-residue escape. The finite stratum, every compact residue window in $(0, 1/2)$, and every sublinear collision are closed. Half-residue escape has an explicit coefficient-growth fingerprint. Consequently every hypothetical bad limit is confined, after parameter-convergent extraction, to a strict first-order Type III or Type IV collision, or to Type IV half-residue escape. A witness-independent normal height must simultaneously tend to infinity, while remaining at most linear in half the Salem degree. These results prove neither Boyd's global closedness conjecture nor the stronger derived-set identity; in particular they do not treat accumulation at 1.

1. Introduction and scope

A Salem number is a real algebraic integer $\tau > 1$ whose conjugates consist of τ^{-1} and points on the unit circle, with at least one unit-circle conjugate. A Pisot number is a real algebraic integer larger than one whose other conjugates lie in the open unit disk. We write $\mathcal{T}$ and $\mathcal{S}$ for the respective sets.

Boyd formulated two related conjectures; see [3, 4]. The closedness conjecture asserts that $\mathcal{S} \cup \mathcal{T}$ is closed in $\mathbf{R}$. The interior sequential component

Date: September 2026.

Key words and phrases. Salem numbers, Pisot numbers, interlacing, Herglotz functions, endpoint compactification.

studied here is the inclusion $\mathcal{T}' \cap (1,\infty) \subseteq \mathcal{S} \cup \mathcal{T}$. Accumulation at 1 is a separate Salem-gap problem, implied in particular by Lehmer's conjecture. Boyd's stronger derived-set conjecture is $(\mathcal{S} \cup \mathcal{T})' = \mathcal{S}$. We prove neither global conjecture. We instead localize every possible failure of the displayed interior component within a specified interlacing-witness subsequence.

During revision we became aware of Amara [2], whose 2025 preprint claims that every accumulation point of $\mathcal{T}$ belongs to $\mathcal{S}$, together with related closedness and Lehmer conclusions. As of 4 September 2026 the arXiv record lists only its initial version and no journal reference. We do not use or assess that manuscript, and nothing below depends on the status of its announced conclusions. The present paper is an independent quantitative compactification and endpoint-localization result, not a claim to settle either Boyd conjecture.

McKee and Smyth's Theorem 7.3 [6] gives every Salem number at least one witness of Types I–IV; the witness need not be unique. Types I and II are governed by CC and CS interlacing, whereas Types III and IV are the type-1 and type-2 SS cases. Types III and IV contain moving exterior poles, yet possess rigid compactifications whose possible failures occur only at explicit endpoint strata. The emphasis on Type IV is also natural in view of [6, Thm. 9.2]: if a least Salem number exists, then it is of Type IV and of no other type. McKee and Smyth already identified Pisot limits for specific limit-function constructions [6, Thms. 4.2 and 5.3]; our concern is different, namely sequences for which the witness itself varies.

Our arguments combine three simple ideas. First, after reversing the interlacing quotient, the Taylor series at the origin has integer coefficients. Second, Herglotz representations yield normal families and uniform coefficient bounds. Third, Cantor's rationality theorem for integer power series of bounded characteristic [5] turns a coefficientwise limit back into a rational function. Meromorphic Hurwitz then transports the unique output zero unless it collides with a pole.

The main results are Theorems 5.3, 6.2, 6.5, and 8.1. Theorem 3.1 removes Type II by an exact algebraic reduction rather than an asymptotic argument. Theorem 7.2 provides a complementary closure principle independent of the chosen McKee–Smyth witness.

For reference, the logical output of the paper is summarized as follows.

chart	proved closure or remaining degeneration
Type I	interior closure
Type II	exact reduction to a closed CC output equation
Type III	closed off collision and in bounded degree; otherwise an unbounded-degree Salem twin with residue comparable to the gap
Type IV	finite-separation and interior-residue strata closed; only first-order collision or half-residue escape remains
normal height	bounded height is closed; every bad sequence has $\kappa \to \infty$ and $\kappa = O(\deg R)$

2. Analytic and arithmetic compactness

We retain the terminology and endpoint conventions of [6, Secs. 2.2–2.5 and Thm. 7.3]. Thus CC means cyclotomic–cyclotomic interlacing, CS means cyclotomic–Salem interlacing, and SS means Salem–Salem interlacing, with the usual caveat that the interlacing polynomials need not themselves be cyclotomic or minimal Salem polynomials. A Type I, II, III, or IV witness uses, respectively, a CC, CS, type-1 SS, or type-2 SS quotient Q/P, where $P, Q \in \mathbf{Z}[z]$ are coprime, monic, of the same degree, and

$$\frac{Q(z)}{(z-1)P(z)} = \frac{2}{z+1} \tag{1}$$

has as its nonzero solutions the output Salem orbit and possibly roots of unity. For $\nu \in \{\mathrm{I}, \mathrm{II}, \mathrm{III}, \mathrm{IV}\}$, let $\mathcal{T}_\nu$ be the Salem outputs admitting a witness of Type ν. These classes need not be disjoint.

For a polynomial F of degree d, put

$$F^*(t) = t^d F(t^{-1}).$$

The reversed quotient associated with an interlacing witness is written

$$\Phi(t) = \frac{Q^*(t)}{P^*(t)}. \tag{2}$$

With monic integral input, $P^*(0) = Q^*(0) = 1$, so $\Phi \in \mathbf{Z}[[t]]$. For the original Type I, III, and IV charts, the output equation has the form

$$G(t) := (1+t)\Phi(t) - 2(1-t) = 0, \tag{3}$$

and the Salem output corresponds to the unique zero $t = \tau^{-1}$ in $\mathbb{D}$. The CC chart obtained from Type II has a different equation, recorded in Theorem 2.5 below.

We use the following input from the interlacing classification. The endpoint alternatives for CS pairs, the real Cauchy transforms, and the signs of their residues are those of [6, Secs. 2.3–2.5 and Thm. 7.3]. For a monic CC pair, the reversed quotient is a Carathéodory function:

$$\Phi(0) = 1, \qquad \Re\Phi(t) > 0 \quad (t \in \mathbb{D}), \qquad |[t^n]\Phi| \le 2 \quad (n \ge 1). \tag{4}$$

Indeed the real partial fractions have positive residues on $[-2, 2]$; lifting their atoms to $\mathbb{T}$ gives the Herglotz representation

$$\Phi(t) = \int_{\mathbb{T}} \frac{\zeta + t}{\zeta - t}\, d\mu(\zeta), \qquad \mu(\mathbb{T}) = 1.$$

For Type III the same partial-fraction description has one positive exterior reciprocal pair, with the type-1 SS ordering $1 < \rho < \tau$. For Type IV it has exactly two negative exterior residues and a positive symmetric compact part. These facts supply the decompositions used in (15) and (27); all estimates after those decompositions are proved here.

We use the following standard arithmetic form of Cantor's theorem.

Theorem 2.1 (Cantor's bounded-characteristic theorem)**.** *Let F be holomorphic in a neighborhood of 0, with Taylor coefficients in $\mathbf{Z}$. If this germ admits a meromorphic continuation of bounded characteristic to $\mathbb{D}$, then $F \in \mathbf{Q}(t)$. In reduced form one may write*

$$F = A/B, \qquad A, B \in \mathbf{Z}[t], \qquad B(0) = 1.$$

Proof. The rationality assertion is Cantor's theorem [5]. For the integral normalization, write $F = \sum a_n t^n$ and consider the Hankel columns $v_n = (a_n, a_{n+1}, \ldots)$. Rationality makes their $\mathbf{Q}$-span finite dimensional. A finite coordinate projection is injective on that span, so the subgroup generated by the v_n embeds in some $\mathbf{Z}^r$ and is finitely generated. The shift $v_n \mapsto v_{n+1}$ is an endomorphism of this lattice. Cayley–Hamilton therefore gives a monic integral recurrence for the a_n. Its generating-function denominator lies in $\mathbf{Z}[t]$ and has constant term one; the numerator is then integral as well. □

Lemma 2.2 (Integer-germ compactness)**.** *Suppose $\Phi_j \in \mathbf{Z}[[t]]$ are meromorphic on $\mathbb{D}$. Assume that there is $\delta > 0$ such that every Φ_j is holomorphic on $|t| < \delta$, the family is normal there, and all moving poles lie outside that disk. Assume also that every fixed Taylor coefficient is bounded and that a subsequential meromorphic limit on pole-free compacta has bounded characteristic. Then a further subsequence converges coefficientwise to an integer series Φ, the meromorphic limit represents that series near zero, and $\Phi \in \mathbf{Q}(t)$.*

Proof. Diagonal extraction makes every integer coefficient eventually constant. Local uniform convergence on $|t| < \delta$ identifies those stabilized coefficients with the derivatives of the meromorphic limit at zero. The limit germ therefore has integer coefficients, and Theorem 2.1 applies. □

If $\Phi = A/B$ is reduced, then at a pole $p \in \mathbb{D}$ the two cleared numerators used below satisfy

$$\begin{aligned} H_{\mathrm{I}}(p) &= (1+p)A(p) \neq 0, \\ H_{3.1}(p) &= 2pA(p) \neq 0, \end{aligned} \tag{5}$$

where

$$H_{\mathrm{I}} = (1+t)A - 2(1-t)B, \qquad H_{3.1} = 2tA - (1-t^2)B.$$

Here $p \neq -1$, $p \neq 0$ because $B(0) = 1$, and $(A, B) = 1$. Thus zeros cannot be hidden at denominator poles. We apply meromorphic Hurwitz only on exhaustions avoiding those poles, together with explicit output–pole separation.

Lemma 2.3 (Integral reversal criterion)**.** *Let $H \in \mathbf{Z}[t]$ satisfy $H(0) = -1$. If H has exactly one zero $u \in (0, 1)$ in $\mathbb{D}$, counted with multiplicity, then $u^{-1} \in \mathcal{S} \cup \mathcal{T}$.*

Proof. If $m = \deg H$, then $-z^m H(z^{-1})$ is monic integral, has $u^{-1} > 1$ as a root, and all its other roots in the closed unit disk. The irreducible factor containing u^{-1} is therefore a Pisot or Salem minimal polynomial. □

Proposition 2.4 (Type I closure)**.** *The Type I output class is closed in the interior in the sense that*

$$\overline{\mathcal{T}_{\mathrm{I}}} \cap (1,\infty) \subseteq \mathcal{S} \cup \mathcal{T}.$$

Proof. Let $\tau_j \to \theta > 1$ and choose monic CC witnesses. By (4), their integer reversed quotients Φ_j form a normal family and every coefficient belongs to a fixed finite set. A subsequence converges locally uniformly to an integer Carathéodory germ Φ, which is rational by Theorem 2.1; write it as A/B in the integral normalized form. The functions in (3) have the unique disk zeros $u_j = \tau_j^{-1} \to \theta^{-1}$. Hurwitz, applied on every compact subdisk, shows that the limit has exactly that one disk zero. The first identity in (5) shows that the cleared polynomial H_{I} has the same disk zeros and $H_{\mathrm{I}}(0) = -1$. Lemma 2.3 finishes the proof. □

Theorem 2.5 (Interior closure for the CC output equation)**.** *Let U_j be monic integral polynomials and C_j integral polynomials of the same degree such that $(2C_j)/U_j$ is a positive CC-interlacing quotient. Put*

$$\Psi_j = C_j^*/U_j^*, \qquad k_j = \Psi_j(0) \in \mathbf{Z}_{>0}.$$

Suppose $u_j = \tau_j^{-1}$ is the unique disk solution of

$$2t\Psi_j(t) = 1 - t^2 \tag{6}$$

and $\tau_j \to \theta > 1$. Then $\theta \in \mathcal{S} \cup \mathcal{T}$.

Proof. The McKee–Smyth numerator is $2C_j$, while Ψ_j is its half-normalized reversed quotient. Thus Ψ_j is Carathéodory of mass k_j. Harnack's inequality on the positive radius and (6) give

$$k_j \frac{1-u_j}{1+u_j} \le \Psi_j(u_j) = \frac{1-u_j^2}{2u_j}, \qquad k_j \le \frac{(1+u_j)^2}{2u_j}.$$

Thus the positive integer masses are bounded, and every fixed Taylor coefficient is bounded by twice that mass. Herglotz compactness and Theorem 2.1 give a rational integer-germ limit $\Psi = A/B$. Hurwitz preserves the unique zero θ^{-1}. By the second identity in (5), the cleared polynomial $H_{3.1} = 2tA - (1-t^2)B$ has exactly that disk zero and satisfies $H_{3.1}(0) = -1$. Apply Lemma 2.3. □

3. The exact Type II reduction

We include Type II because it removes an apparent endpoint family before the Type III/IV analysis begins. Let $P, Q \in \mathbf{Z}[z]$ be a monic CS-interlacing pair of the same degree producing a Type II output. Thus the nonunit solutions of

$$\frac{Q(z)}{(z-1)P(z)} = \frac{2}{z+1} \tag{7}$$

are the prescribed Salem root, its conjugates, and allowed unit roots.

Theorem 3.1 (CS-to-CC Weyl complement)**.** *Put*

$$U = \frac{Q}{z-1}, \qquad C = \frac{(z+1)Q - (z-1)P}{z}. \tag{8}$$

By the CS endpoint alternatives in [6, Sec. 2.3]*, if* $r = \operatorname{ord}_{z=1} Q$*, then* $r \in \{1, 3\}$*. Define*

$$g = \begin{cases} 1, & r = 1, \\ z-1, & r = 3, \end{cases} \qquad \widetilde{U} = U/g, \quad \widetilde{C} = C/g.$$

Then $2\widetilde{C}/\widetilde{U}$ *is an admissible CC quotient for the McKee–Smyth Theorem 3.1 construction, and*

$$2z\widetilde{C} - (z^2-1)\widetilde{U} = \frac{(z+1)Q - 2(z-1)P}{g}. \tag{9}$$

Thus at most the cyclotomic factor $z-1$ *is removed, every nonunit output root is preserved, and*

$$\mathcal{T}_{\mathrm{II}} \subseteq \mathcal{T}_{3.1}.$$

Here $\mathcal{T}_{3.1}$ *denotes the output class of the CC construction in* [6, Thm. 3.1]*. In particular,*

$$\overline{\mathcal{T}_{\mathrm{II}}} \cap (1, \infty) \subseteq \mathcal{S} \cup \mathcal{T}. \tag{10}$$

Proof. The CS endpoint convention makes the multiplicity of 1 in Q odd and at most three. Since $P(0) = 1$ and $Q(0) = -1$, the second numerator in (8) is divisible by z; its leading terms cancel, so C is integral of degree at most $\deg Q - 1$, while U is monic. Reciprocity shows that U is reciprocal and C antireciprocal.

In the real-line variable, write the CS quotient as a positive-residue Cauchy transform $m(x) \sim x^{-1}$. Its reciprocal Weyl complement

$$L(x) = x - \frac{1}{m(x)}$$

has positive residue $-1/m'(\beta)$ at every zero β of m, all of which lie in $[-2, 2]$, and $L = O(x^{-1})$. Hence it is again a positive Cauchy transform with circularly interlacing numerator and denominator. Its total mass

$$k = \lim_{x \to \infty} xL(x) > 0$$

is the leading coefficient of C, so $k \in \mathbf{Z}_{>0}$ and $\deg C = \deg U$. Direct substitution identifies it with the transform of C/U. Scaling the numerator by 2 does not change its zeros or the CC interlacing, so $2C/U$ is a CC quotient as well. The only common factor is $z-1$, absent for $r = 1$ and simple after cancellation for $r = 3$. Clearing the transformed equation gives (9).

For completeness, let $a > 2$ be the largest CS pole. The CS ordering gives $m(x) < 0$ on $(2, a)$, so

$$L(2+) = 2 - \frac{1}{m(2+)} > 2$$

(with value $+\infty$ at an endpoint zero). Thus the real transform of $2\widetilde{C}/\widetilde{U}$ satisfies the endpoint condition in [6, Thm. 3.1]. If $\Psi = \widetilde{C}^*/\widetilde{U}^*$, reversing (9) at a nonunit root gives exactly

$$2t\Psi(t) = 1 - t^2.$$

This explains both the factor 2 in the McKee–Smyth numerator and the half-normalized choice of Ψ in Theorem 2.5. Theorem 2.5 therefore proves the asserted interior closure. □

4. Why the Type II descent does not iterate

The Weyl complement in Theorem 3.1 is geometrically tempting in Type III as well. The next proposition shows that its arithmetic normalization can fail at every depth; this is why the direct integer-germ compactness argument in the following section is needed.

Proposition 4.1 (Failure of fixed-lattice Weyl iteration)**.** *There is a monic Type III witness whose first Weyl complement is again a positive-residue Cauchy transform, but whose second normalized complement has Laurent coefficients with unbounded powers of 2 in their denominators. Consequently no fixed nonzero rational scaling puts all of those coefficients in* $\mathbf{Z}$*.*

Proof. Consider

$$P(z) = z^6 - 2z^5 - z^4 - 5z^3 - z^2 - 2z + 1,$$
$$Q(z) = z^6 - 2z^5 + 2z - 1.$$

Its real interlacing quotient is

$$m(x) = \frac{x(x^4 - 6x^2 + 7)}{(x^2 - 1)(x^2 - x - 3)(x^2 + x - 3)}. \tag{11}$$

The numerator zeros are

$$-\sqrt{3+\sqrt{2}},\ -\sqrt{3-\sqrt{2}},\ 0,\ \sqrt{3-\sqrt{2}},\ \sqrt{3+\sqrt{2}},$$

and they strictly alternate with the six displayed denominator zeros. The largest denominator zero $(1+\sqrt{13})/2$ is larger than the largest numerator zero, so this is the type-1 SS ordering. Moreover

$$(z+1)Q - 2(z-1)P = -(z-1)(z^2+1)(z^4 - 4z^3 - z^2 - 4z + 1). \tag{12}$$

The last factor has trace polynomial $X^2 - 4X - 3$ and is irreducible by Eisenstein at 7 after replacing z by $z+1$. Thus the pair is a genuine Type III witness.

The first Weyl complement and its total mass are

$$L(x) = x - \frac{1}{m(x)} = \frac{(x^2-3)(2x^2-3)}{x(x^4 - 6x^2 + 7)}, \qquad \lim_{x\to\infty} xL(x) = 2. \tag{13}$$

The positive normalized next step is therefore forced to be

$$M(x) = x - \frac{2}{L(x)} = \frac{x(3x^2-5)}{(x^2-3)(2x^2-3)}. \tag{14}$$

Put $t = x^{-2}$ and write $M = x^{-1}\sum_{n\geq 0} a_n t^n$. Directly from (14),

$$(2 - 9t + 9t^2)\sum_{n\geq 0} a_n t^n = 3 - 5t.$$

If $b_n = 2^{n+1}a_n$, then $b_0 = 3$, $b_1 = 17$, and

$$b_n = 9b_{n-1} - 18b_{n-2} \qquad (n \geq 2).$$

Every b_n is odd, so $v_2(a_n) = -(n+1)$. For a fixed $q \in \mathbf{Q}^\times$, the valuations $v_2(qa_n)$ are therefore negative for all sufficiently large n. No fixed scaling restores an integral Laurent lattice. □

5. Type III: noncollision closure

Fix one Type III witness for each output τ_j. In the disk its reversed quotient has one exterior pole pair. Let $1 < \rho_j < \tau_j$ be the exterior root of the denominator and let $\lambda_j > 0$ be the residue assigned to each member of that reciprocal pair. Put $u_j = \tau_j^{-1}$.

Proposition 5.1 (Type III decomposition and sharpened budgets)**.** *There is a real-symmetric Carathéodory function C_j such that*

$$\Phi_j(t) = C_j(t) + \frac{2\lambda_j(1-t^2)}{(1-\rho_j t)(1-\rho_j^{-1}t)}, \qquad C_j(0) = 1 - 2\lambda_j \geq 0. \tag{15}$$

At the output zero define

$$\delta_j = \frac{(1+u_j)C_j(u_j)}{2(1-u_j)}.$$

Then $0 \leq \delta_j < 1$ and the exact defect factorization is

$$\lambda_j = \frac{(\tau_j - \rho_j)(\rho_j\tau_j - 1)}{\rho_j(\tau_j+1)^2}(1-\delta_j). \tag{16}$$

In particular both bounds

$$\lambda_j \leq \frac{(\tau_j - \rho_j)(\rho_j\tau_j - 1)}{\rho_j(\tau_j+1)^2}, \tag{17}$$

$$\lambda_j \leq \frac{(\tau_j - \rho_j)(\rho_j\tau_j - 1)}{2\tau_j(\rho_j+1)^2} \tag{18}$$

hold globally.

Proof. Remove the two positive exterior atoms from the positive Cauchy measure of the Type III quotient; the remaining symmetric atoms lie on $[-2,2]$ and give C_j. Their masses add to $1 - 2\lambda_j$, proving (15). Put

$$D_j(u) = (1-\rho_j u)(1-\rho_j^{-1}u), \qquad h(u) = \frac{2(1-u)}{1+u}.$$

The exact output equation is

$$h(u_j) - C_j(u_j) = \frac{2\lambda_j(1-u_j^2)}{D_j(u_j)} > 0. \tag{19}$$

Solving this identity and substituting $u_j = \tau_j^{-1}$ gives (16); it also shows $0 \le \delta_j < 1$ and gives (17).

Harnack's inequality on the positive radius gives

$$C_j(u_j) \ge C_j(0)\frac{1-u_j}{1+u_j} = (1-2\lambda_j)\frac{1-u_j}{1+u_j}.$$

Combining this with (19) yields

$$2\lambda_j\big((1+u_j)^2 - D_j(u_j)\big) \le D_j(u_j).$$

Since

$$(1+u_j)^2 - D_j(u_j) = \frac{(\rho_j+1)^2}{\rho_j\tau_j},$$

the second bound follows. □

Lemma 5.2 (Simplicity of the Type III output)**.** *The disk output zero of a Type III witness is simple.*

Proof. In the real variable the positive-residue odd Cauchy transform has the form

$$m(x) = \frac{\eta_0}{x} + \sum_{a>0} \eta_a \left(\frac{1}{x-a} + \frac{1}{x+a}\right), \qquad \eta_a > 0,$$

and the output equation is $xm(x) = 2$. To the right of the largest pole,

$$\frac{d}{dx}(xm(x)) = -4x \sum_{a>0} \frac{\eta_a a^2}{(x^2-a^2)^2} < 0,$$

where strictness follows from the exterior pole. The disk change of variable is locally nonsingular at the output, proving the claim. □

Theorem 5.3 (Type III noncollision closure)**.** *Let $\tau_j \to \theta > 1$ be Type III Salem outputs with chosen witnesses. If*

$$\liminf_j (\tau_j - \rho_j) > 0, \tag{20}$$

then $\theta \in \mathcal{S} \cup \mathcal{T}$. Within the compactification (15), the only way for the output zero to be lost is

$$\tau_j - \rho_j \longrightarrow 0, \qquad \lambda_j \longrightarrow 0. \tag{21}$$

Every ρ_j is a Salem number or a reciprocal-quadratic Pisot number.

Proof. After a subsequence, $\rho_j \to \rho \ge 1$, $\lambda_j \to \lambda$, and $C_j \to C$ locally uniformly. Each fixed Taylor coefficient of (15) is bounded uniformly in j. Moreover $\rho_j < \tau_j = O(1)$, so the disk poles ρ_j^{-1} stay outside a fixed neighborhood $|t| < \delta$ of zero. The limit is a Carathéodory function plus one rational pole pair and hence has bounded characteristic. Since the full germs

are integral, Lemma 2.2 gives a rational limit $\Phi = A/B$ in the normalized integral form of Theorem 2.1.

Under (20), u_j remains separated from the moving pole ρ_j^{-1}. Choose a closed disk about $u = \theta^{-1}$ disjoint from the limiting pole set. For large j, G_j is holomorphic there and has its one simple zero u_j there, so the argument principle gives one limit zero counted with multiplicity. A zero elsewhere on a pole-free compact disk would, by Hurwitz, force a second disk zero of G_j. The first identity in (5) excludes a zero created by clearing a reduced denominator pole. Hence H_{I} has exactly the one disk zero u, and Lemma 2.3 gives $\theta \in \mathcal{S} \cup \mathcal{T}$.

It remains to identify the only loss mode. If $\rho > 1$ and $\lambda = 0$, put

$$D_j = (1 - \rho_j t)(1 - \rho_j^{-1} t), \qquad K_j = D_j G_j.$$

Then K_j is holomorphic, its only disk zero is u_j, and

$$K_j(\rho_j^{-1}) = 2\lambda_j(1 + \rho_j^{-1})(1 - \rho_j^{-2}) \neq 0.$$

But

$$K_j \longrightarrow (1 - \rho t)(1 - \rho^{-1} t)\big[(1 + t)C - 2(1 - t)\big],$$

which is not identically zero and vanishes at ρ^{-1}. Divisor-form Hurwitz forces $u_j \to \rho^{-1}$, i.e. $\tau_j - \rho_j \to 0$. Conversely, (17) makes every collision residue tend to zero. If $\rho = 1$, the limiting pole $\rho^{-1} = 1$ lies on $\partial\mathbb{D}$ and is separated from $u = \theta^{-1}$. If $\lambda > 0$, the limiting form of (17) forces $\theta - \rho > 0$, so again $u < \rho^{-1}$. In either case meromorphic Hurwitz retains the output zero; the same cleared polynomial H_{I}, the coprime-pole check (5), and Lemma 2.3 then rule out any further loss mode.

Finally the monic denominator has only the reciprocal pair ρ_j, ρ_j^{-1} off the unit circle. Its noncyclotomic irreducible factor is therefore Salem, or reciprocal quadratic Pisot. □

Corollary 5.4 (Closed Type III strata)**.** *The limit belongs to $\mathcal{S} \cup \mathcal{T}$ if either*

(i) $\liminf_j \lambda_j > 0$, *or*
(ii) *the degrees $[\mathbf{Q}(\rho_j) : \mathbf{Q}]$ are bounded.*

Consequently a genuinely bad Type III sequence must satisfy

$$\rho_j \in \mathcal{T}, \quad [\mathbf{Q}(\rho_j) : \mathbf{Q}] \to \infty, \quad \rho_j, \tau_j \to \theta, \quad \lambda_j \to 0. \tag{22}$$

Proof. On a compact τ-range, (17) bounds λ_j by a fixed multiple of $\tau_j - \rho_j$; a positive residue lower bound therefore implies separation. For bounded degree, $1 < \rho_j < \tau_j = O(1)$ and the Salem/Pisot property give bounded absolute Weil height. Northcott finiteness [7] leaves finitely many ρ_j. If collision occurs, a constant subsequence has $\theta = \rho \in \mathcal{S} \cup \mathcal{T}$; otherwise apply Theorem 5.3. □

The information discarded at collision has a finite, nonzero first derivative.

Proposition 5.5 (Strict first collision tangent)**.** *Assume the collision regime, pass to a subsequence for which $C_j \to C$ locally uniformly on $\mathbb{D}$, and suppose $\rho_j, \tau_j \to \theta > 1$. Then*

$$\frac{\lambda_j}{\tau_j - \rho_j} \longrightarrow L_{\theta,C} := \frac{1}{2\theta}\left(\frac{2(\theta-1)}{\theta+1} - C(\theta^{-1})\right), \tag{23}$$

and

$$0 < L_{\theta,C} \le \frac{\theta-1}{2\theta(\theta+1)}. \tag{24}$$

In particular

$$\lambda_j = L_{\theta,C}(\tau_j - \rho_j) + o(\tau_j - \rho_j).$$

Proof. Put $u_j = \tau_j^{-1}$ and $r_j = \rho_j^{-1}$. Solving (19) gives the exact identity

$$\frac{\lambda_j}{\tau_j - \rho_j} = \frac{u_j(1 - r_j u_j)}{2(1-u_j^2)}\left(\frac{2(1-u_j)}{1+u_j} - C_j(u_j)\right). \tag{25}$$

Local uniform convergence and $u_j, r_j \to r := \theta^{-1}$ yield (23).

It remains to show that the limiting defect is nonzero. Set

$$D_j(t) = (1-\rho_j t)(1-\rho_j^{-1}t), \qquad K_j = D_j G_j.$$

The simple pole of G_j at r_j is cancelled and

$$K_j(r_j) = 2\lambda_j(1+r_j)(1-r_j^2) \neq 0.$$

By Lemma 5.2, K_j has exactly one disk zero counted with multiplicity, namely u_j. On compact subsets of $\mathbb{D}$,

$$K_j \longrightarrow K(t) := (1-\theta t)(1-\theta^{-1}t)\big((1+t)C(t) - 2(1-t)\big).$$

The first factor gives a zero of K at r. On a small closed disk about r whose boundary is zero-free, the argument principle shows that this zero has multiplicity one. Since $1-\theta t$ already has a simple zero there,

$$(1+r)C(r) - 2(1-r) \neq 0.$$

The bracket in (25) is positive before passage to the limit; nonvanishing makes its limit strictly positive. Hence $L_{\theta,C} > 0$.

Finally $\lambda_j \to 0$ gives $C(0) = 1$, and Harnack's inequality yields

$$C(r) \ge \frac{1-r}{1+r} = \frac{\theta-1}{\theta+1}.$$

Substitution in (23) proves (24). □

Corollary 5.6 (Uniform first-order Type III collision)**.** *Along every Type III collision with $\tau_j \to \theta > 1$,*

$$0 < \liminf_j \frac{\lambda_j}{\tau_j - \rho_j} \le \limsup_j \frac{\lambda_j}{\tau_j - \rho_j} \le \frac{\theta-1}{2\theta(\theta+1)}.$$

Thus $\lambda_j \asymp \tau_j - \rho_j$; the sublinear collision $\lambda_j = o(\tau_j - \rho_j)$ is impossible.

Proof. The upper bound follows from (18). If the lower limit were zero, choose a subsequence on which the ratio tends to zero and then a further locally convergent Carathéodory subsequence. Proposition 5.5 would give a strictly positive limit, a contradiction. □

The right side depends on the limiting Carathéodory function, not only on θ. A useful algebraic model for the loss of an integer germ is obtained from $A = z^2 - z - 1$, $A^* = 1 - z - z^2$, and

$$P_k = \frac{z^k A - A^*}{z - 1}, \qquad Q_k = (z-1)P_k.$$

Direct reversal gives

$$Q_k^* - (1-t)P_{k+1}^* = t^k(t-1)A(t).$$

Thus every fixed Taylor prefix stabilizes while the zero of A can remain visible only at the moving scale. We use this identity solely as a coefficient-compactness diagnostic; no admissibility claim for these pairs is needed in any theorem of the paper.

6. Type IV: residue budget and compactification

Use throughout the Type IV real variable

$$x = \sqrt{z} + \frac{1}{\sqrt{z}}. \tag{26}$$

For a chosen Type IV witness its real quotient has the exact signed Cauchy form

$$\mathcal{R}(x) = \int_{[-2,2]} \frac{d\nu(y)}{x-y} - \frac{c}{x-a} - \frac{c}{x+a}, \qquad a > 2, \quad c > 0, \tag{27}$$

where ν is a positive symmetric atomic measure and

$$\nu([-2,2]) - 2c = 1. \tag{28}$$

The output point $x_\tau = \sqrt{\tau} + \tau^{-1/2}$ lies in $(2, a)$ and obeys

$$\mathcal{R}(x_\tau) = \frac{2}{x_\tau}. \tag{29}$$

Proposition 6.1 (Exact Type IV defect and universal budget)**.** *For a Type IV witness put*

$$\Delta = \int_{[-2,2]} \frac{y^2}{x_\tau^2 - y^2}\, d\nu(y).$$

Then $0 \le \Delta < 1$ *and*

$$\Delta + \frac{2ca^2}{a^2 - x_\tau^2} = 1, \qquad c = \frac{a^2 - x_\tau^2}{2a^2}(1-\Delta). \tag{30}$$

Consequently

$$c \le \frac{a^2 - x_\tau^2}{2a^2} < \frac{1}{2}, \qquad \|\nu - c\delta_a - c\delta_{-a}\|_{\mathrm{TV}} = 1 + 4c < 3. \tag{31}$$

Moreover

$$1-2c=\Delta+\frac{2cx_\tau^2}{a^2-x_\tau^2},\qquad \int_{[-2,2]} y^2\,d\nu(y)\le x_\tau^2(1-2c). \tag{32}$$

Consequently

$$|\mathcal{R}(z)|<\frac{3}{|\operatorname{Im} z|}\qquad (z\notin\mathbf{R}). \tag{33}$$

Proof. For $x>2$, symmetry gives

$$\int\frac{d\nu(y)}{x-y}=\int\frac{x\,d\nu(y)}{x^2-y^2}.$$

At $x=x_\tau<a$, the exterior pair contributes $2cx/(a^2-x^2)$. Multiplying the output equation by x_τ and using (28) gives exactly

$$2=1+2c+\Delta+\frac{2cx_\tau^2}{a^2-x_\tau^2}=1+\Delta+\frac{2ca^2}{a^2-x_\tau^2},$$

which proves (30); because $c>0$, it also gives $\Delta<1$. Rearrangement gives the first identity in (32). Since

$$\frac{y^2}{x_\tau^2-y^2}\ge\frac{y^2}{x_\tau^2},$$

the moment bound follows. The total variation is $\nu([-2,2])+2c=1+4c$, and the Cauchy-kernel bound gives (33). □

Consider a sequence $\tau_j\to\theta>1$ with chosen Type IV witnesses and write $x_j=x_{\tau_j}\to x_0>2$. By the budget, after passing to a subsequence,

$$a_j\to a\in[x_0,\infty],\qquad c_j\to c\in[0,1/2],\qquad \nu_j\Rightarrow\nu.$$

The Type IV ordering makes x_j the unique zero in $(2,a_j)$ of $H_j(x)=x\mathcal{R}_j(x)-2$, and

$$H_j(x)>0\qquad (x_j<x<a_j). \tag{34}$$

Theorem 6.2 (Exact Type IV compactification)**.** *Every subsequential limit belongs to exactly one of the following strata.*

(i) Finite separated: $x_0<a<\infty$, *and necessarily* $0<c<1/2$.
(ii) Zero-residue collision: $a=x_0$ *and* $c_j\to0$.
(iii) Half-residue escape: $a_j\to\infty$, *and necessarily*

$$c_j\to\frac12,\qquad \nu_j\Rightarrow2\delta_0,\qquad \mathcal{R}_j\longrightarrow\frac{2}{x} \tag{35}$$

locally uniformly on $\mathbb{C}\setminus[-2,2]$.

Proof. The budget immediately turns $a_j-x_j\to0$ into $c_j\to0$. Suppose now $x_0<a<\infty$ but $c=0$. Set

$$J(x)=x\int_{[-2,2]}\frac{d\nu(y)}{x-y}=\int_{[-2,2]}\frac{x^2}{x^2-y^2}\,d\nu(y).$$

Its derivative is

$$J'(x) = -2x \int_{[-2,2]} \frac{y^2}{(x^2 - y^2)^2}\, d\nu(y) \le 0. \tag{36}$$

Separation lets us pass to the output equation, giving $J(x_0) = 2$, while (34) gives $J(x) \ge 2$ for $x_0 < x < a$. Monotonicity gives the reverse inequality, so $J \equiv 2$ on an interval. Formula (36) forces ν to be supported at zero, and $J = 2$ then gives $\nu = 2\delta_0$, contradicting (28), which would give mass one when $c = 0$. Hence $c > 0$.

If $a_j \to \infty$, the exterior terms vanish at every fixed $x > 2$. The same one-sided argument gives $J(x_0) = 2$ and $J(x) \ge 2$ for every $x > x_0$, hence $J \equiv 2$ and $\nu = 2\delta_0$. The mass identity then gives $c = 1/2$. These alternatives exhaust the convergent subsequences. □

The escape is singular at the arithmetic base point. Write

$$a_j = \sqrt{\eta_j} + \eta_j^{-1/2}, \qquad \eta_j > 1.$$

The disk decomposition is

$$\Phi_j(t) = C_j(t) - \frac{2c_j(1 - t^2)}{(1 - \eta_j t)(1 - \eta_j^{-1} t)}, \qquad C_j(0) = 1 + 2c_j < 2. \tag{37}$$

In the escape regime $\eta_j^{-1} \to 0$ and

$$\Phi_j(t) \longrightarrow \frac{2(1 - t)}{1 + t} \tag{38}$$

locally uniformly on compact subsets of $\mathbb{D} \setminus \{0\}$, whereas $\Phi_j(0) = 1$ and the right side has value 2 at zero. The integer germ is literally lost at the puncture. Indeed $\nu_j \Rightarrow 2\delta_0$ in the real Cauchy chart, and the Cauchy-to-disk transform used in deriving (37) sends this atom to

$$C_j(t) \longrightarrow \frac{2(1 - t)}{1 + t}$$

locally in $\mathbb{D}$. For fixed $t \neq 0$, the exterior term in (37) tends to zero because its denominator is asymptotic to $-\eta_j t$. At $t = 0$, however, that term equals $-2c_j \to -1$, while $C_j(0) \to 2$, leaving $\Phi_j(0) = 1$.

The puncture has a coefficient-level arithmetic signature.

Corollary 6.3 (Coefficient fingerprint of Type IV escape)**.** *Write*

$$\Phi_j(t) = \sum_{n \ge 0} \phi_{n,j} t^n, \qquad C_j(t) = \sum_{n \ge 0} \gamma_{n,j} t^n.$$

In the half-residue escape regime, for each fixed $n \ge 1$,

$$\phi_{n,j} = \gamma_{n,j} - 2c_j(\eta_j^n + \eta_j^{-n}), \qquad \frac{\phi_{n,j}}{\eta_j^n} \longrightarrow -1, \qquad \frac{\phi_{n,j}}{a_j^{2n}} \longrightarrow -1. \tag{39}$$

In particular every fixed nonconstant integer Taylor coefficient tends to $-\infty$.

Proof. The coefficient of t^n in

$$\frac{1-t^2}{(1-\eta t)(1-\eta^{-1}t)}$$

is $\eta^n+\eta^{-n}$ for $n \geq 1$. Herglotz's coefficient bound gives $|\gamma_{n,j}| \leq 2(1+2c_j) < 4$. Now use $c_j \to 1/2$, $\eta_j \to \infty$, and $a_j^2/\eta_j \to 1$. □

Theorem 6.4 (Type IV nondegenerate closure)**.** *If*

$$\sup_j a_j < \infty, \qquad \liminf_j (a_j - x_j) > 0, \tag{40}$$

then $\theta \in \mathcal{S} \cup \mathcal{T}$.

Proof. Theorem 6.2 places the limit in the finite separated stratum and keeps $c > 0$. Formula (37), the bounds $C_j(0) < 2$, and bounded η_j make every fixed Taylor coefficient bounded. The same bound puts the disk pole η_j^{-1} outside one fixed neighborhood of zero. The signed-Herglotz limit has bounded characteristic, so Lemma 2.2 gives a rational integer-germ limit $\Phi = A/B$. Output–pole separation in (40) and the local nonsingularity of $s \mapsto \sqrt{s} + s^{-1/2}$ separate the disk zero $u_j = \tau_j^{-1}$ uniformly from the pole $p_j = \eta_j^{-1}$. On a pole-avoiding exhaustion, the argument principle retains the one disk zero and Hurwitz excludes any additional one. The first coprimality check in (5) excludes a zero created at a reduced denominator pole. Thus H_{I} has the same unique disk zero and satisfies $H_{\mathrm{I}}(0) = -1$. Lemma 2.3 finishes the proof. □

Theorem 6.5 (Sublinear Type IV collision closure)**.** *Let* $\tau_j \to \theta > 1$ *be Type IV outputs with chosen witnesses, and put*

$$x_j = \sqrt{\tau_j} + \tau_j^{-1/2}, \qquad x_0 = \sqrt{\theta} + \theta^{-1/2}.$$

If

$$a_j - x_j \longrightarrow 0, \qquad \frac{c_j}{a_j - x_j} \longrightarrow 0, \tag{41}$$

then $\theta \in \mathcal{S} \cup \mathcal{T}$.

Proof. Write $a_j = \sqrt{\eta_j} + \eta_j^{-1/2}$ with $\eta_j > 1$. The defining map is strictly increasing on $(1, \infty)$, so collision gives $\eta_j \to \theta$, while (31) gives $c_j \to 0$. After extraction, $\nu_j \Rightarrow \nu$ with $\nu([-2,2]) = 1$. Put

$$\mathcal{C}_j(x) = \int_{[-2,2]} \frac{d\nu_j(y)}{x-y}, \qquad \mathcal{C}(x) = \int_{[-2,2]} \frac{d\nu(y)}{x-y}.$$

At the output,

$$\mathcal{C}_j(x_j) = \frac{2}{x_j} - \frac{2x_j}{a_j + x_j} \frac{c_j}{a_j - x_j}.$$

The second term tends to zero, and the Cauchy kernels converge uniformly for x near $x_0 > 2$. Hence $\mathcal{C}(x_0) = 2/x_0$.

Let C_j and C be the disk Carathéodory transforms of ν_j and ν. Then $C_j \to C$ locally uniformly in $\mathbb{D}$. By (39), with $c_j \to 0$ and bounded η_j, every

fixed Taylor coefficient of the exterior term in (37) tends to zero. Since $\Phi_j \in \mathbf{Z}[[t]]$, every Taylor coefficient of C is an integer. The Carathéodory function C has bounded characteristic, so Theorem 2.1 gives $C = A/B \in \mathbf{Q}(t)$ with $A, B \in \mathbf{Z}[t]$ and $B(0) = 1$.

Put $u_j = \tau_j^{-1}$, $u_0 = \theta^{-1}$, and

$$G_j(t) = (1+t)\Phi_j(t) - 2(1-t), \qquad G(t) = (1+t)C(t) - 2(1-t).$$

The moving pole $p_j = \eta_j^{-1}$ also tends to u_0. On every compact subset of $\mathbb{D} \setminus \{u_0\}$ the exterior term tends uniformly to zero, so $G_j \to G$ there. Each G_j has no disk zero other than u_j. Because $G(0) = -1$, the zero-free form of Hurwitz shows that G has no zero in $\mathbb{D} \setminus \{u_0\}$.

For $t \in (0,1)$ set $x(t) = t^{-1/2} + t^{1/2}$. The real and disk transforms are related by

$$C(t) = \frac{1-t}{\sqrt{t}}\, \mathcal{C}(x(t)),$$

and hence

$$G(t) = (1-t)\big(J(x(t)) - 2\big), \qquad J(x) = x\mathcal{C}(x) = \int_{[-2,2]} \frac{x^2}{x^2 - y^2}\, d\nu(y).$$

The output identity gives $G(u_0) = 0$. Moreover

$$J'(x_0) = -2x_0 \int_{[-2,2]} \frac{y^2}{(x_0^2 - y^2)^2}\, d\nu(y) < 0.$$

Equality would force $\nu = \delta_0$, contrary to $J(x_0) = 2$. Since $x'(u_0) \neq 0$ and $u_0 \neq 1$, the zero of G at u_0 is simple.

The reduced denominator B is zero-free in $\mathbb{D}$, since C is holomorphic there. Therefore

$$H = (1+t)A - 2(1-t)B$$

has exactly the one simple disk zero u_0 and satisfies $H(0) = -1$. Lemma 2.3 yields $\theta \in \mathcal{S} \cup \mathcal{T}$. □

Corollary 6.6 (Residue-window closure and endpoint rates)**.** *For every $0 < \varepsilon \le 1/4$, Type IV closure holds on the stratum*

$$\varepsilon \le c_j \le \frac{1}{2} - \varepsilon. \tag{42}$$

Every individual witness satisfies the quantitative separation

$$a - x_\tau \ge x_\tau \left((1-2c)^{-1/2} - 1 \right). \tag{43}$$

For any sequence with $x_j \to x_0 > 2$,

$$c_j \to 0 \iff a_j - x_j \to 0, \qquad c_j \to \frac{1}{2} \iff a_j \to \infty.$$

If $a_j - x_j \to 0$ and $\nu_j \Rightarrow \nu$, put

$$\Delta_0 = \int_{[-2,2]} \frac{y^2}{x_0^2 - y^2}\, d\nu(y).$$

Then the exact collision slope is

$$\frac{c_j}{a_j - x_j} \longrightarrow \frac{1 - \Delta_0}{x_0} \leq \frac{1}{x_0}. \tag{44}$$

If $x_0 > 2\sqrt{2}$, it also obeys

$$\frac{x_0^2 - 8}{x_0(x_0^2 - 4)} \leq \lim_j \frac{c_j}{a_j - x_j} \leq \frac{1}{x_0}. \tag{45}$$

Finally, if $\theta \notin \mathcal{S} \cup \mathcal{T}$, every collision sequence satisfies

$$0 < \liminf_j \frac{c_j}{a_j - x_j} \leq \limsup_j \frac{c_j}{a_j - x_j} \leq \frac{1}{x_0}. \tag{46}$$

The only other endpoint is the forced escape (35)–(38).

Proof. The budget gives $x_\tau/a \leq \sqrt{1 - 2c}$, which is (43). It also shows that collision implies $c_j \to 0$. Conversely, if $c_j \to 0$ but the gaps do not tend to zero, an unbounded pole subsequence would force $c_j \to 1/2$, while a bounded-pole subsequence would have a finite separated limit with $c > 0$. Both contradict Theorem 6.2. The equivalence at $1/2$ follows by the same compactification argument.

The two bounds in (42) now exclude collision and escape, respectively, so Theorem 6.4 applies. In a collision, let Δ_j be the defect in Proposition 6.1. Weak convergence and $x_j \to x_0 > 2$ give $\Delta_j \to \Delta_0$, while (30) gives

$$\frac{c_j}{a_j - x_j} = \frac{a_j + x_j}{2a_j^2}(1 - \Delta_j).$$

This proves (44). In collision the limiting measure has mass one; hence

$$\Delta_0 \leq \frac{4}{x_0^2 - 4}.$$

This yields (45) when $x_0 > 2\sqrt{2}$.

If a bad collision had zero lower slope, a subsequence would satisfy the hypotheses of Theorem 6.5, contradicting $\theta \notin \mathcal{S} \cup \mathcal{T}$. The upper limit follows directly by dividing (31) by $a_j - x_j$. □

7. A witness-independent normal-height constraint

Endpoint localization can be sharpened independently of the choice of a McKee–Smyth witness. Let $R \in \mathbf{Z}[z]$ be a Salem minimal polynomial of degree $2s$, and write uniquely

$$R(z) = z^s r(z + z^{-1}), \qquad r \in \mathbf{Z}[X]. \tag{47}$$

The polynomial r is monic irreducible of degree s: monicity and integrality follow from reciprocal descent, while a proper factor of r would lift to a proper factor of the Salem minimal polynomial R. Its roots are the exterior trace $y = \tau + \tau^{-1} > 2$ and $s - 1$ simple roots in $(-2, 2)$. Since a Salem

minimal polynomial has no cyclotomic factor, zero is not a root of r. Thus the roots of $f = Xr$ are

$$\alpha_1 < \cdots < \alpha_s \quad (-2 < \alpha_i < 2), \qquad y > 2,$$

where one of the compact roots α_i is the added root $X = 0$.

Definition 7.1 (Normal height)**.** A strict integral normal interlacer is a polynomial $b \in \mathbf{Z}[X]$ of degree $s-1$, with negative leading coefficient and one simple zero in each compact gap (α_i, α_{i+1}). Set

$$\kappa(R) = \min_b(-\operatorname{lc} b). \tag{48}$$

The minimum is well defined. Indeed, choose rational points $\xi_i \in (\alpha_i, \alpha_{i+1})$ and clear denominators in $-\prod_{i=1}^{s-1}(X - \xi_i)$. The possible values of $-\operatorname{lc} b$ form a nonempty subset of $\mathbf{Z}_{>0}$. A minimizing interlacer is primitive, since division by a nontrivial content would lower its leading height while preserving its zeros.

Theorem 7.2 (Bounded normal-height closure)**.** *Let R_j be Salem minimal polynomials with roots $\tau_j \to \theta > 1$. If $\kappa(R_j)$ is bounded, then $\theta \in \mathcal{S} \cup \mathcal{T}$.*

Proof. Pass to a subsequence with a fixed leading coefficient $c = -\kappa < 0$ for a minimizing interlacer b_j. If $b_j = c\prod_{k=1}^{s_j-1}(X - \xi_{k,j})$ with $\alpha_{k,j} < \xi_{k,j} < \alpha_{k+1,j}$, direct sign counting gives positive residues at every $\alpha_{i,j}$ and a negative residue at y_j. Since $b_j/f_j = O(X^{-2})$ at infinity, the sum of all residues is zero. We may therefore write uniquely

$$\frac{b_j(X)}{f_j(X)} = \sum_{i=1}^{s_j} \frac{\lambda_{i,j}}{X - \alpha_{i,j}} - \frac{\Lambda_j}{X - y_j}, \qquad \lambda_{i,j}, \Lambda_j > 0, \quad \sum_i \lambda_{i,j} = \Lambda_j. \tag{49}$$

Because f_j is monic integral and $\deg f_j - \deg b_j = 2$,

$$\frac{b_j(X)}{f_j(X)} \in X^{-2}\mathbf{Z}[[X^{-1}]].$$

Thus all signed power moments are integers. For the integer Chebyshev trace polynomials

$$C_0 = 2, \quad C_1 = X, \quad C_{n+1} = XC_n - C_{n-1},$$

the signed moments

$$a_{j,n} = \sum_i \lambda_{i,j} C_n(\alpha_{i,j}) - \Lambda_j C_n(y_j) \tag{50}$$

the signed moments are therefore integers as well. For fixed n,

$$|a_{j,n}| \le \Lambda_j(2 + C_n(y_j)), \tag{51}$$

which is uniformly bounded by (52) below and $y_j \to \theta + \theta^{-1}$.

Let $\bar{\alpha}_j = \Lambda_j^{-1}\sum_i \lambda_{i,j}\alpha_{i,j} \in [-2, 2]$. The first moment is $c = \Lambda_j(\bar{\alpha}_j - y_j)$, so

$$\frac{|c|}{y_j + 2} \le \Lambda_j \le \frac{|c|}{y_j - 2}. \tag{52}$$

Because $\theta > 1$, the masses are bounded above and away from zero. Lift $\alpha_{i,j} = \zeta_{i,j} + \zeta_{i,j}^{-1}$ to the unit circle and form the symmetric positive measure

$$\mu_j = \frac{1}{2} \sum_i \lambda_{i,j} (\delta_{\zeta_{i,j}} + \delta_{\zeta_{i,j}^{-1}}).$$

After weak-* and diagonal extraction, $\Lambda_j \to \Lambda > 0$ and every integer $a_{j,n}$ stabilizes to a_n. The Herglotz function

$$U(t) = 2 \int_{\mathbb{T}} \frac{d\mu(\zeta)}{1 - t\zeta}$$

has positive real part. Hence $(U-1)/(U+1) \in H^\infty$, so U has bounded characteristic. The function

$$\mathcal{A}(t) = \sum_{n \geq 0} a_n t^n = U(t) - \frac{\Lambda}{1 - \theta t} - \frac{\Lambda}{1 - \theta^{-1} t} \tag{53}$$

extends meromorphically with bounded characteristic to $\mathbb{D}$. Theorem 2.1 makes $\mathcal{A}$ rational. The term $-\Lambda/(1-\theta t)$ gives a genuine pole at θ^{-1}, because U and the other rational term are holomorphic there and $\Lambda > 0$. It is the only pole in $\mathbb{D}$. If A/B is the reduced integral form with $B(0) = 1$, then

$$B^\vee(z) = z^{\deg B} B(z^{-1})$$

is monic integral, has the unique root θ outside the closed unit disk, and has every other root in that disk. The irreducible factor containing θ makes it Pisot or Salem. □

There is also a uniform linear upper bound, so a hypothetical bad sequence is confined to a linear-height cusp rather than an arbitrary one.

Theorem 7.3 (Linear normal-height bound)**.** *For a Salem minimal polynomial of degree $2s$ and Salem root τ,*

$$\kappa(R) < (16\tau^2 + 48)(s - 1). \tag{54}$$

Proof. Let C_m be as above, put

$$H_m = 2 + C_m, \qquad h_m = (2 - X) H_m, \qquad D_m = (y - 2) H_m(y),$$

where $C_m(y) = \tau^m + \tau^{-m}$. Thus $D_m \to \infty$, so the least $m \geq 0$ with $D_m > 16(s-1)$ exists; recall also that $s \geq 2$ for a Salem polynomial. Define the integral remainder

$$b_m = \operatorname{rem}_r \left[X r'(X) h_m(X) \right]. \tag{55}$$

Because r is monic, Euclidean division preserves integrality, so $b_m \in \mathbf{Z}[X]$ and $\deg b_m < s$. For a compact conjugate $\beta \in (-2, 2)$, one has $0 < H_m(\beta) \leq 4$; if $H_m(\beta) = 0$, irreducibility would force r to divide H_m, impossible at y. Hence

$$0 < h_m(\beta) < 16, \qquad |\beta h_m(\beta)| < 32, \qquad h_m(y) = -D_m.$$

At each root γ of r, $b_m(\gamma)/(\gamma r'(\gamma)) = h_m(\gamma)$. The residue at every compact pole is positive and the exterior residue is negative. Lagrange interpolation at the roots of the monic polynomial r gives the two exact trace identities

$$\frac{b_m(0)}{r(0)} = -\operatorname{Tr}_{\mathbf{Q}(y)/\mathbf{Q}} h_m > D_m - 16(s-1) > 0, \tag{56}$$

$$[X^{s-1}]b_m = \operatorname{Tr}_{\mathbf{Q}(y)/\mathbf{Q}}(Xh_m), \qquad \left|[X^{s-1}]b_m + yD_m\right| < 32(s-1). \tag{57}$$

Since $y > 2$ and $D_m > 16(s-1)$, the second estimate in (57) also gives $[X^{s-1}]b_m < -yD_m + 32(s-1) < 0$. Thus $\deg b_m = s-1$ and this coefficient is $\operatorname{lc} b_m$. The first identity is the residue at the added zero pole $X = 0$. Thus all compact residues of b_m/f are positive and the exterior residue is negative. The function crosses from one infinite sign to the other in every compact gap; since $\deg b_m = s-1$, it has exactly one simple zero in every such gap and is a strict normal interlacer.

If $m \geq 1$, minimality gives $D_{m-1} \leq 16(s-1)$. Since $\tau C_{m-1}(y) - C_m(y) = \tau^{-m}(\tau^2 - 1) > 0$ and $2 < 2\tau$, one has

$$D_m < \tau D_{m-1} \leq 16\tau(s-1).$$

For $m = 0$, use $y - 2 = (\tau-1)^2/\tau < \tau$ and $s \geq 2$ to get $D_0 = 4(y-2) < 4\tau \leq 16\tau(s-1)$. Combining this with (57) and $y\tau = \tau^2 + 1$ gives

$$\kappa(R) \leq -\operatorname{lc} b_m < (16y\tau + 32)(s-1) = (16\tau^2 + 48)(s-1).$$

□

Corollary 7.4 (Exact computation at fixed input)**.** *For every fixed Salem minimal polynomial R, the integer $\kappa(R)$ is effectively computable by a finite exact search.*

Proof. Put $n = s-1$ and $B_R = (16\tau^2 + 48)n$. Theorem 7.3 reduces the leading height to $1 \leq K \leq \lceil B_R \rceil - 1$; the ceiling is decidable by exact real-root isolation. If

$$b = -K\prod_{i=1}^{n}(X - \xi_i)$$

is a strict normal interlacer, then $|\xi_i| < 2$, so Vieta's formulas give

$$\left|[X^{n-q}]b\right| \leq K\binom{n}{q}2^q \qquad (0 \leq q \leq n).$$

Only finitely many integral candidates occur. Sturm sequences and exact root isolation decide whether a candidate has one simple zero in every prescribed compact gap. Searching increasing K returns $\kappa(R)$. □

8. Aggregate endpoint localization

Recall the four output classes defined in Section 2, always with one witness selected for each term. Type I is closed by Proposition 2.4, and Type II is closed by Theorem 3.1.

Theorem 8.1 (Three-cusp localization)**.** *Suppose $\tau_j \in \mathcal{T}$ and $\tau_j \to \theta > 1$ with $\theta \notin \mathcal{S} \cup \mathcal{T}$. Choose one McKee–Smyth witness for every j, pass to a subsequence of fixed type, and then, if necessary, to a further subsequence on which the pole, residue, and compact-measure parameters converge. Exactly one of the following descriptions holds for the resulting subsequence. In the Type IV alternatives, put*

$$x_j = \sqrt{\tau_j} + \tau_j^{-1/2}, \qquad x_0 = \sqrt{\theta} + \theta^{-1/2}.$$

(I) Type III Salem-twin collision:

$$\rho_j \in \mathcal{T}, \quad [\mathbf{Q}(\rho_j) : \mathbf{Q}] \to \infty, \quad \rho_j, \tau_j \to \theta, \quad \tau_j - \rho_j \to 0, \quad \lambda_j \to 0$$

$$0 < \liminf_j \frac{\lambda_j}{\tau_j - \rho_j} \le \limsup_j \frac{\lambda_j}{\tau_j - \rho_j} \le \frac{\theta - 1}{2\theta(\theta + 1)}.$$

(II) Type IV zero-residue collision:

$$a_j - x_j \to 0, \quad c_j \to 0, \quad 0 < \liminf_j \frac{c_j}{a_j - x_j} \le \limsup_j \frac{c_j}{a_j - x_j} \le \frac{1}{x_0}.$$

(III) Type IV half-residue escape:

$$a_j \to \infty, \quad c_j \to \frac{1}{2}, \quad \nu_j \Rightarrow 2\delta_0, \quad \mathcal{R}_j \to 2/x \text{ locally uniformly on } \mathbb{C}\backslash[-2, 2],$$

with loss of the integer germ at $t = 0$ as in (38).

In every case, if R_j is the Salem minimal polynomial of τ_j and $2s_j = \deg R_j$, then

$$s_j \to \infty, \qquad \kappa(R_j) \to \infty, \qquad \kappa(R_j) < (16\tau_j^2 + 48)(s_j - 1). \tag{58}$$

Proof. There are only four witness types. A fixed-type subsequence cannot be Type I or II by their closure results. Theorem 5.3 and Corollaries 5.4 and 5.6 give (I). Theorems 6.2, 6.4, and 6.5, together with Corollary 6.6, give (II) or (III). These are alternatives only within the resulting parameter-convergent subsequence; a different witness choice or subsequence may produce a different local chart.

If $\kappa(R_j)$ had a bounded subsequence, Theorem 7.2 would put θ in $\mathcal{S} \cup \mathcal{T}$. Thus the height tends to infinity along every subsequence. If s_j were bounded, then $h(\tau_j) = \log \tau_j/(2s_j)$ and the degrees would both be bounded; Northcott finiteness would leave a constant subsequence, contradicting the bad limit. Hence $s_j \to \infty$. The final inequality is Theorem 7.3; since $\tau_j \to \theta$, it also says $\kappa(R_j) = O_\theta(s_j)$. □

9. What remains

Theorem 8.1 converts an amorphous closure problem into three specific endpoint problems, but it does not eliminate them. A complete proof of interior closure must eliminate all three, either through a common arithmetic rigidity principle or through separate mechanisms adapted to the three cases:

- cross-degree rigidity of the strictly positive Type III tangent (23) in unbounded-degree Salem-twin collisions;
- recovery of the cancelled pole in the strictly first-order Type IV zero-residue collision; and
- recovery of an arithmetic germ through the Type IV escape puncture, despite the coefficient growth in (39).

Coefficient stabilization alone does not exclude the first or third regime, as the explicit collision identity and (38) demonstrate.

There is a precise interface with Mahler-function arithmetic that may be useful after a terminal rescaling, but the rescaling itself remains missing.

Proposition 9.1 (Adelic Mahler pole interface)**.** *Let $F \in \mathbf{Q}[[t]]$ be k-Mahler for some fixed $k \geq 2$. Assume that its p-adic Taylor radius is at least one for every rational prime p, and that its complex germ is meromorphic in $\mathbb{D}$ with exactly one pole $\rho \in (0,1)$. Then ρ^{-1} is a Pisot or Salem number.*

Proof. Let $d(t)$ be the normalized Mahler denominator of F in the sense of [1, Def. 3.9]. Galois invariance and normalization give $d \in \mathbf{Q}[t]$. The denominator identity shows that $d(\rho) = 0$: otherwise all iterates $F(t^{k^i})$ are holomorphic at ρ, contradicting the pole. The finite-place part of [1, Prop. 6.4] and the assumed radii show that ρ^{-1} is an algebraic integer. The same local denominator argument at the complex embeddings shows that every other conjugate of ρ^{-1} has modulus at most one, since a second conjugate of ρ strictly inside $\mathbb{D}$ would be a second pole. If all inequalities are strict the number is Pisot. Otherwise irreducibility and a unit-circle conjugate force the minimal polynomial to be reciprocal, leaving precisely the reciprocal pair off the unit circle; the number is Salem. □

Proposition 9.1 is conditional at the Salem–Boyd cusps: this paper constructs neither a fixed-base Mahler equation nor the required all-place analytic germ in any of the three terminal regimes. It therefore supplies a target criterion, not an additional closure claim.

All conclusions in this paper are relative to $\theta > 1$. Even a resolution of the three endpoints would establish only

$$\overline{\mathcal{T}} \cap (1, \infty) \subseteq \mathcal{S} \cup \mathcal{T}.$$

It would not exclude accumulation at 1, and it would not prove the stronger derived-set identity $(\mathcal{S} \cup \mathcal{T})' = \mathcal{S}$.

Verification note

The residue-defect identities, the Type II cleared identity, the two Weyl complements in Proposition 4.1, the collision changes of variables, and the escape coefficients follow by direct expansion from the displayed formulas. No theorem in this paper depends on numerical or computer-algebra evidence. Compactness, rationality, zero retention, and Northcott finiteness are proved or cited in the text.

References

[1] B. Adamczewski, J. P. Bell and D. Smertnig, *A height gap theorem for coefficients of Mahler functions*, J. Eur. Math. Soc. (JEMS) **25** (2023), no. 7, 2525–2571, doi:10.4171/JEMS/1244.

[2] M. Amara, *Nombres de Pisot, nombres de Salem et la conjecture de Lehmer*, arXiv:2509.21402v1 [math.NT] (2025), arXiv:2509.21402.

[3] D. W. Boyd, *Small Salem numbers*, Duke Math. J. **44** (1977), no. 2, 315–328, doi:10.1215/S0012-7094-77-04413-1.

[4] D. W. Boyd, *Pisot and Salem numbers in intervals of the real line*, Math. Comp. **32** (1978), no. 144, 1244–1260, doi:10.1090/S0025-5718-1978-0491587-8.

[5] D. G. Cantor, *Power series with integral coefficients*, Bull. Amer. Math. Soc. **69** (1963), 362–366, doi:10.1090/S0002-9904-1963-10923-4.

[6] J. McKee and C. Smyth, *Salem numbers and Pisot numbers via interlacing*, Canad. J. Math. **64** (2012), no. 2, 345–367, doi:10.4153/CJM-2011-051-2.

[7] D. G. Northcott, *An inequality in the theory of arithmetic on algebraic varieties*, Proc. Cambridge Philos. Soc. **45** (1949), no. 4, 502–509, doi:10.1017/S0305004100025202.